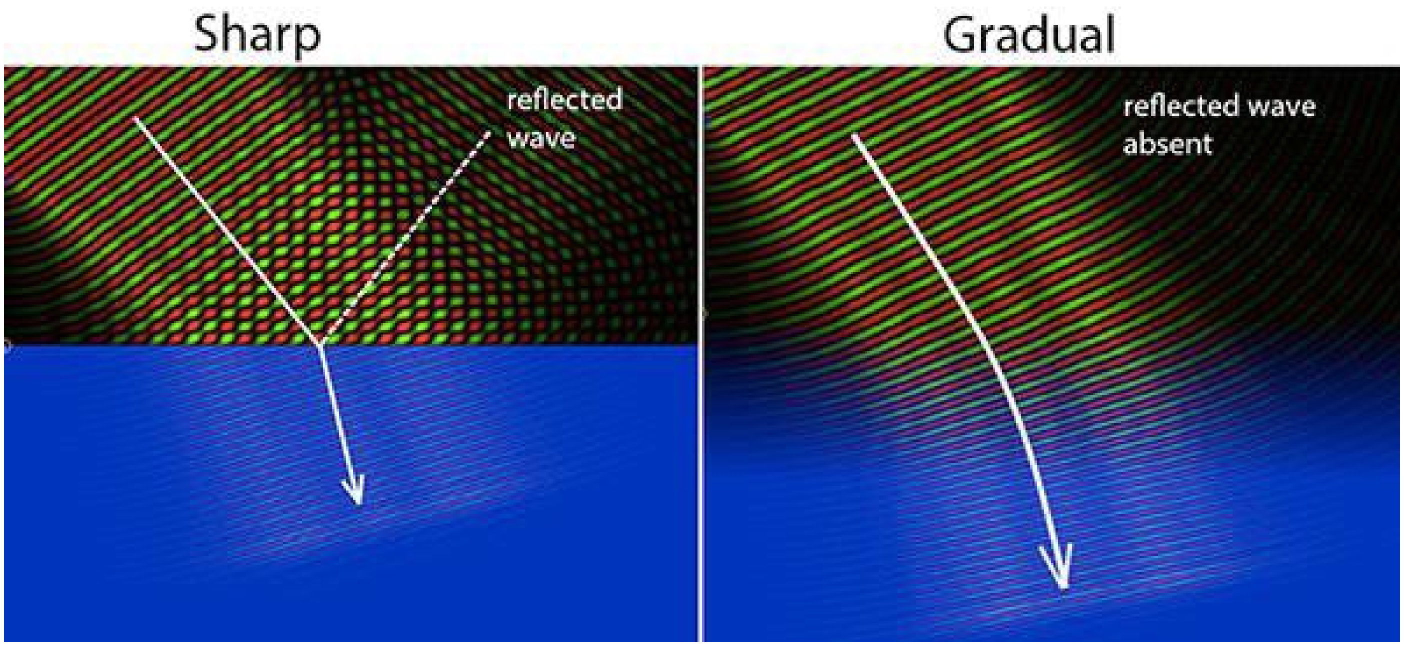
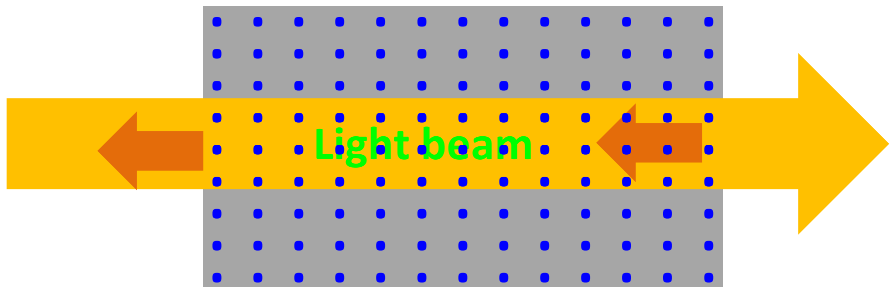
Reflection

- When a beam of light strikes an interface, some light is always scattered backward and we call this phenomenon **Reflection**.



- The transition between two gradual media, there will be very little reflection.

Reflection

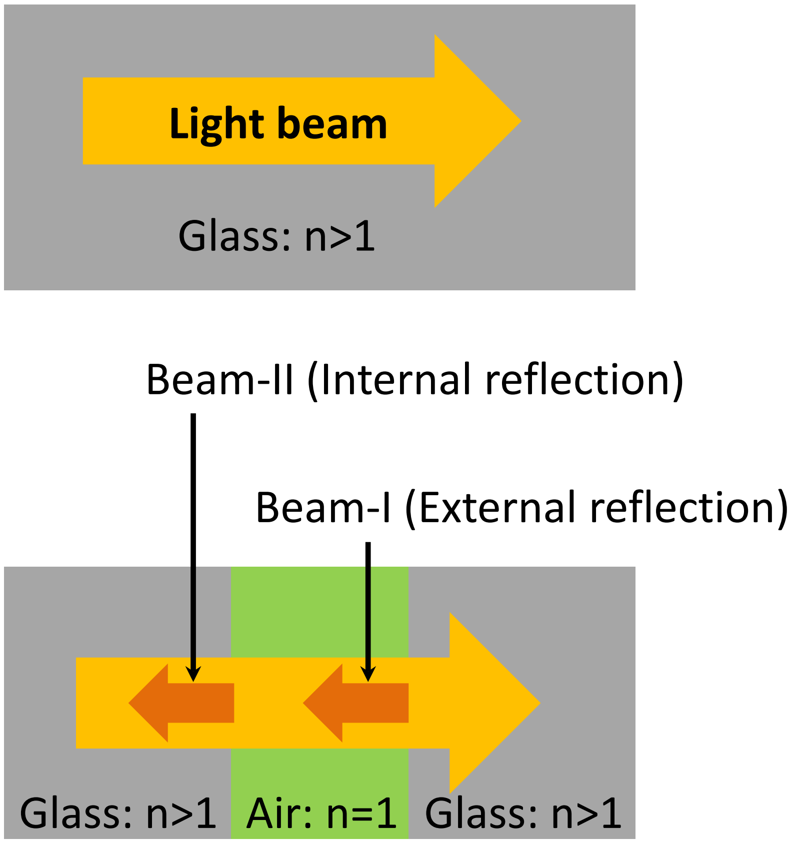


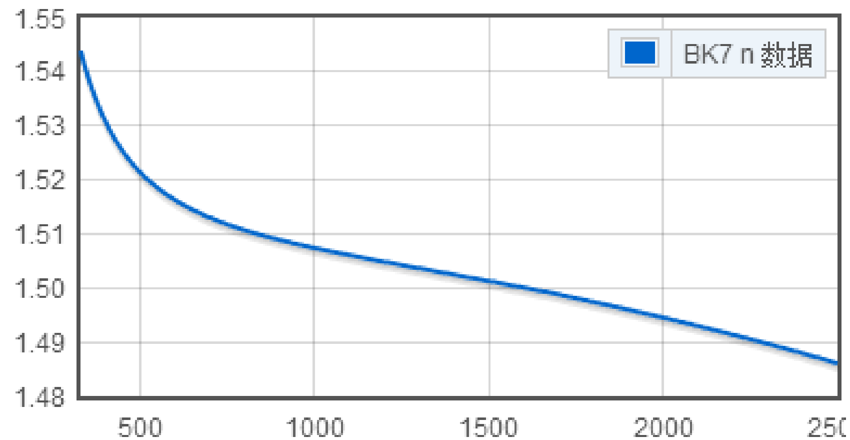
Each molecule scatters light in the backward direction, and in principle, each and every molecule contributes to the reflected wave.

In practice, it is a thin layer (**$\approx\lambda/2$ deep**) of unpaired atomic oscillators near the surface that is effectively responsible for the reflection.

Example: an air-glass interface, about **4%** of the energy of an incident beam falling perpendicularly in air on glass will be reflected straight back out by this layer of unpaired scatters.

Internal and external reflection





- **External reflection**: light is traveling from a less to a more optically dense medium ($n_i<n_t$)
- **Internal reflection**: ($n_i>n_t$)

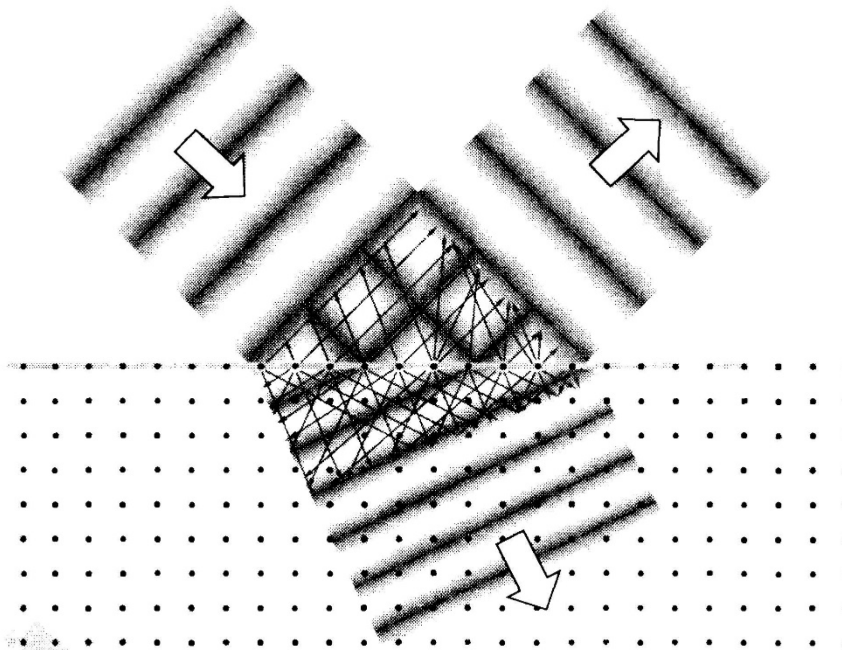
❖ Note: 180° relative phase shift between internally and externally reflected light

White light is reflected as white

- The layer of scatterers responsible for the reflection is effectively about $\lambda/2$ thick. Hence, the larger the wavelength, the deeper the region contributing, and the more scatterers there are acting together.
- Each scatter is less efficient as λ increase ($\propto 1/\lambda^4$)

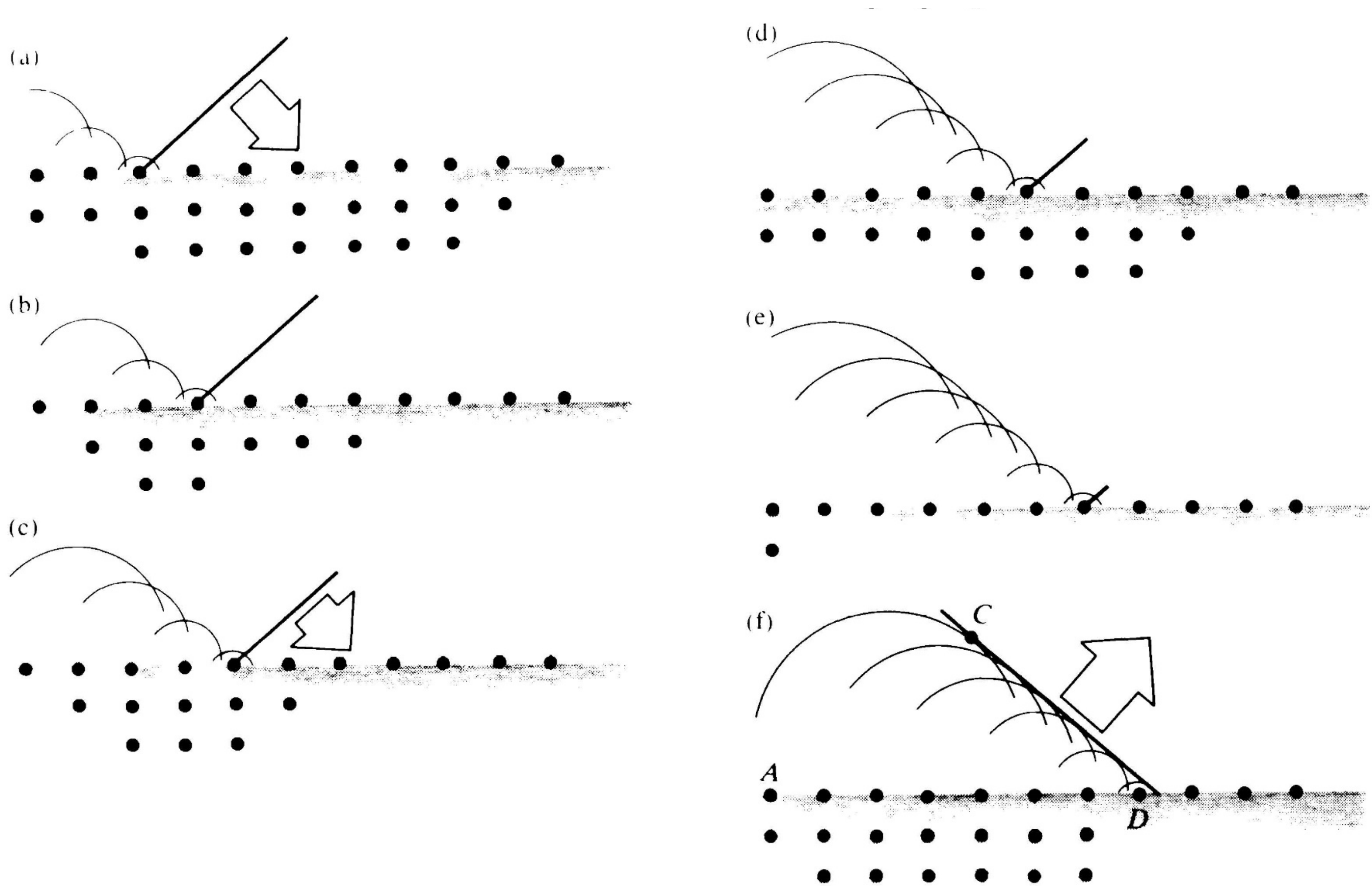
Two effects tend to balance out.

The law of reflection



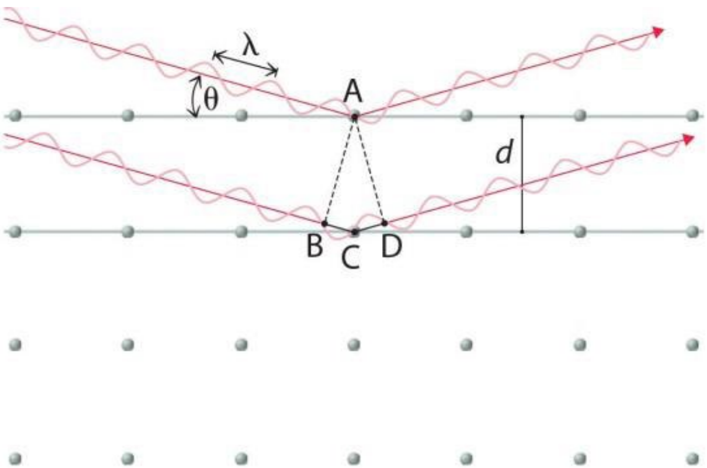
Wavelength >> separation (molecules or atoms):
A plane wave sweeps in stimulating atoms across the interface. These radiate and reradiate, thereby giving rise to both the reflected and transmitted waves

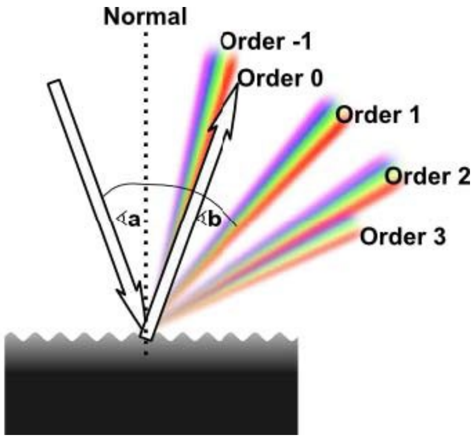
How it happens



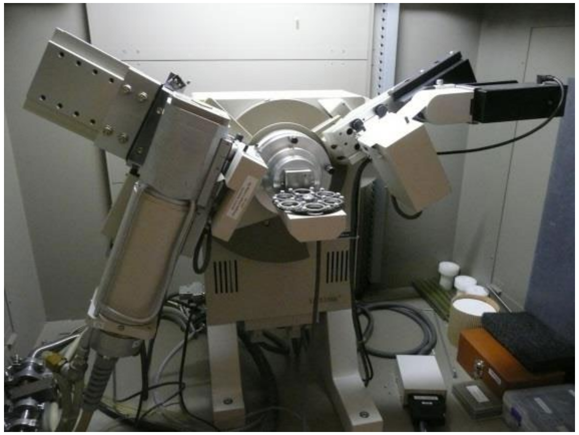
The reflection of a wave as the result of scattering

❑ Wavelength < separation: **diffraction**





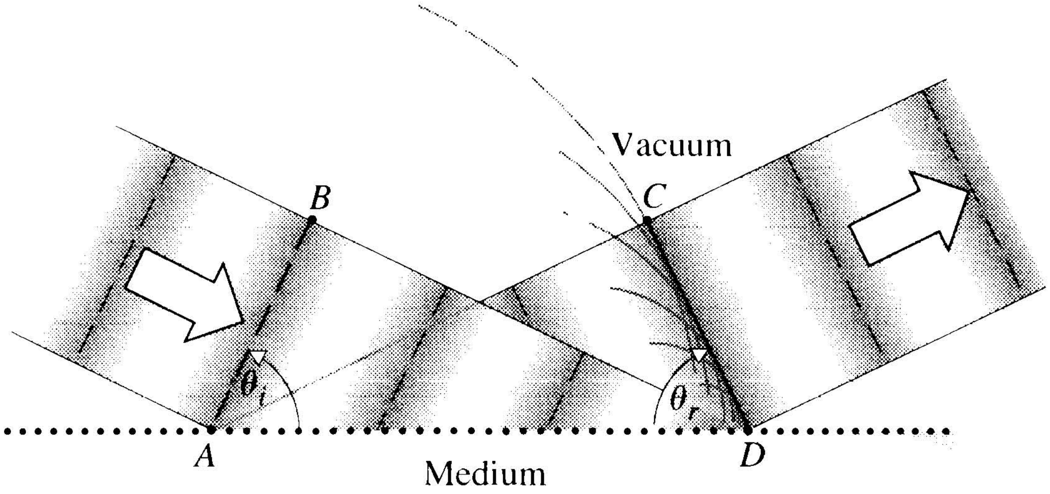
The X-ray diffractometer



Spectrometer

Law of reflection

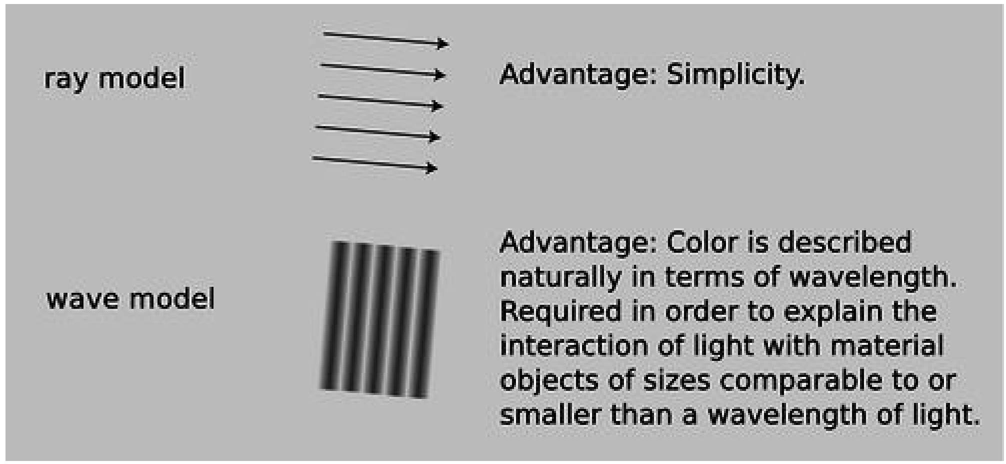
The wavelet emitted from A will arrive at C in-phase with the wavelet just being emitted from D, as long as the distances $\overline{AC}$ and $\overline{BD}$are equal.



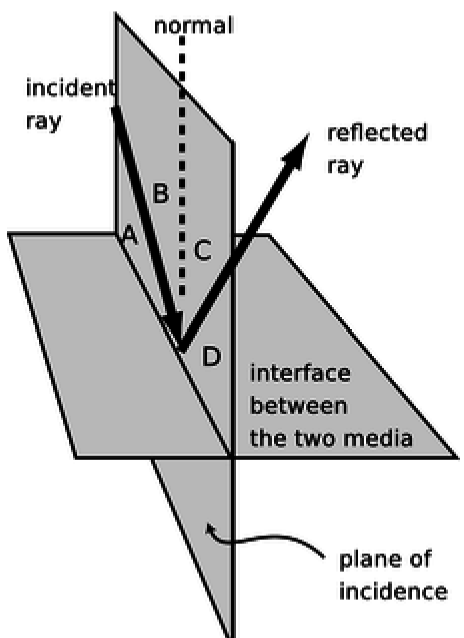
$$\frac{\sin\theta_i}{\overline{BD}} = \frac{\sin\theta_r}{\overline{AC}} \qquad \overline{BD} = v_i \Delta t = \overline{AC}$$

$\theta_i = \theta_r$ ❖ **The angle-of-incidence equals the angle-of-reflection**

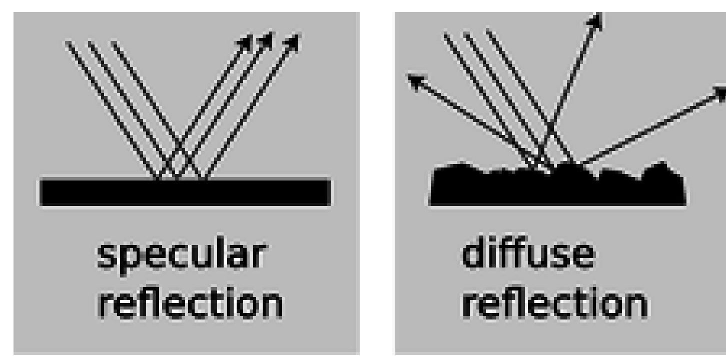
Ray



Plane-of-incidence: the plane where the incident ray, the perpendicular to the surface, and the reflected ray all lie in.



Specular and diffuse reflection

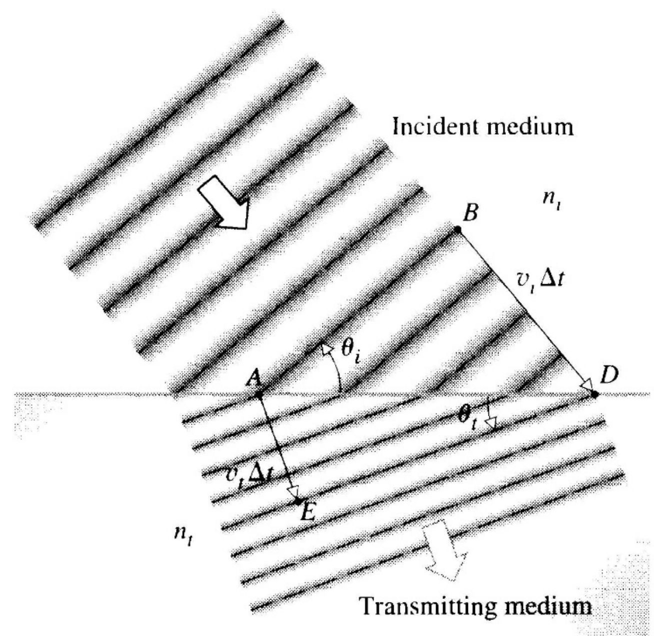


3.4 Refraction

Immediately on entering the transmitting medium, the transmitted wave propagates with an effective speed $v_t < c$ for air-glass interface.

The law of refraction:

In the time Δt, which it takes for point B on a wavefront (traveling at speed v_i) to reach point D, the transmitted portion of that same wavefront (traveling at speed v_t) has reached point E.



$$\frac{\sin\theta_i}{\overline{BD}} = \frac{\sin\theta_t}{\overline{AE}}$$

$$\overline{BD} = v_i \Delta t$$

$$\overline{AE} = v_t \Delta t$$

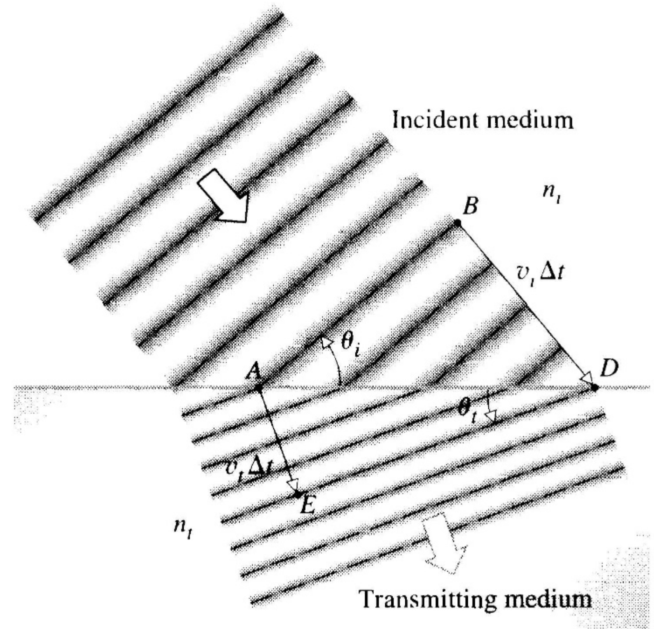
$$\frac{\sin\theta_i}{v_i} = \frac{\sin\theta_t}{v_t}$$

$$n_i \sin\theta_i = n_t \sin\theta_t$$

The law of refraction or Snell's law

Three important changes that occur in the beam traversing the interface:

- ❑1) it changes direction;
- ❑2) the beam in the glass has a broader cross section than the beam in the air;
- ❑3) the wavelength decrease because the frequency is unchanged with the speed decreases.

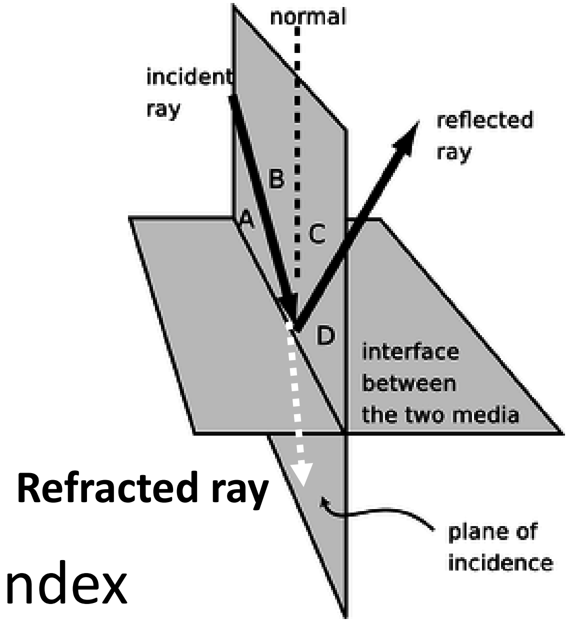


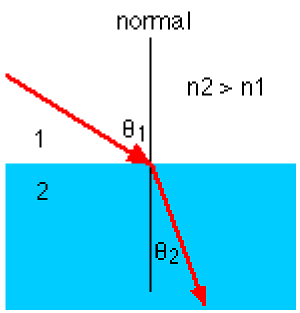
Note: the color aspect of light is better thought of as associated with its frequency (or energy, E=$h\nu$)

When we talk about wavelength and colors, we should always be referring to **vacuum wavelengths**.

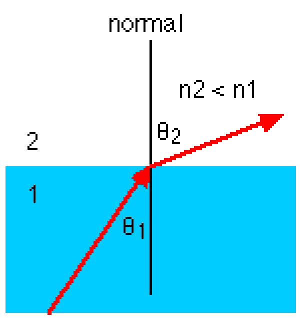
$$\lambda = \frac{v}{\nu} = \frac{c}{n\nu} = \frac{\lambda_0}{n}$$

The incident, reflected, and refracted rays all lie in the plane-of-incidence

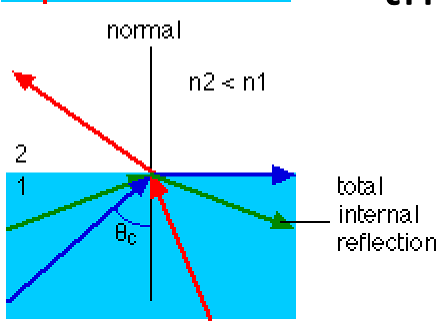




The ray entering a higher-index medium bends toward the normal.



On entering a medium having a lower index, the ray will bend away from the normal.



Total internal reflection

More

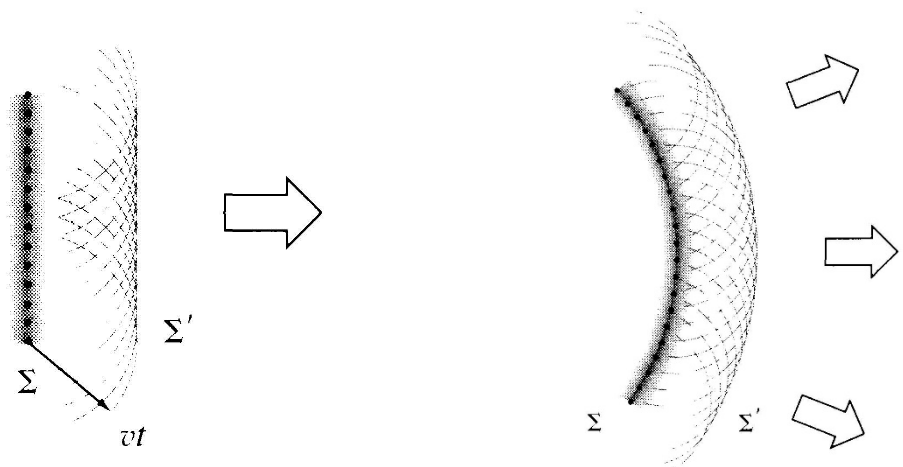
- In all the situations treated thus far, it was assumed that the reflected and refracted beams always had **the same frequency** as the incident beam.
- It is the truth when the amplitude of the vibration is fairly small.
- ✓ E-field for bright sunlight: 1000 V/m
- ✓ Cohesive field holding the electron in an atom: 10^{11} V/m.
- It is not true for exceedingly large-amplitude E-field, for example high power laser (pulse laser)

Nonlinear effect: for example second harmonic generation (SHG)

3.4.2 Huygens' Principle

Every point on a propagating wavefront serves as the source of spherical secondary wavelets, such that the wavefront at some later time is the envelope of these wavelets.

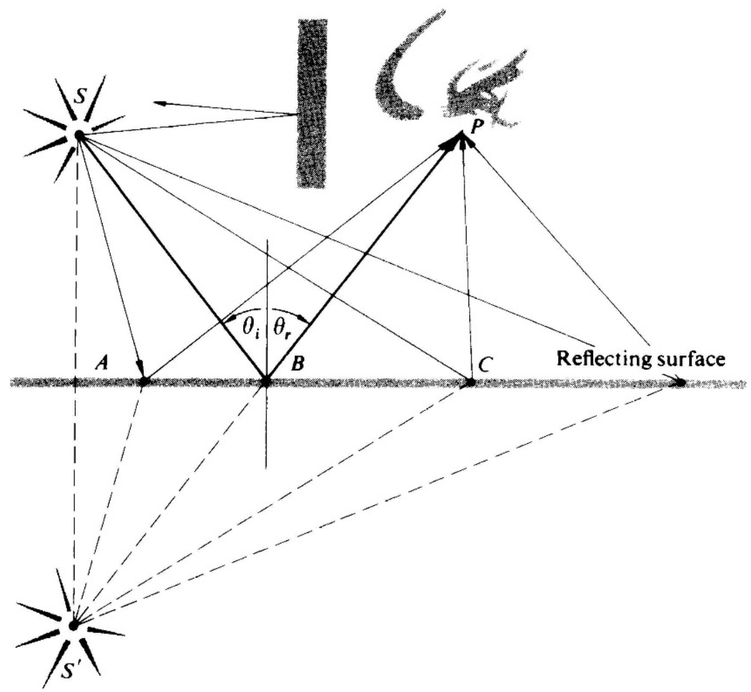
The wavelets themselves are a theoretical construct.



Huygens-Fresnel Principle: Fresnel successfully modified Huygens' Principle mathematically adding in the concept of interference.

3.5 Fermat's principle

Hero's statement: "the path taken by light in going from some point S to a point P via a reflecting surface was the shortest possible one "



Fermat consequently reformulated Hero's statement to read: The actual path between two points taken by a beam of light is the one that is traversed in the least time.

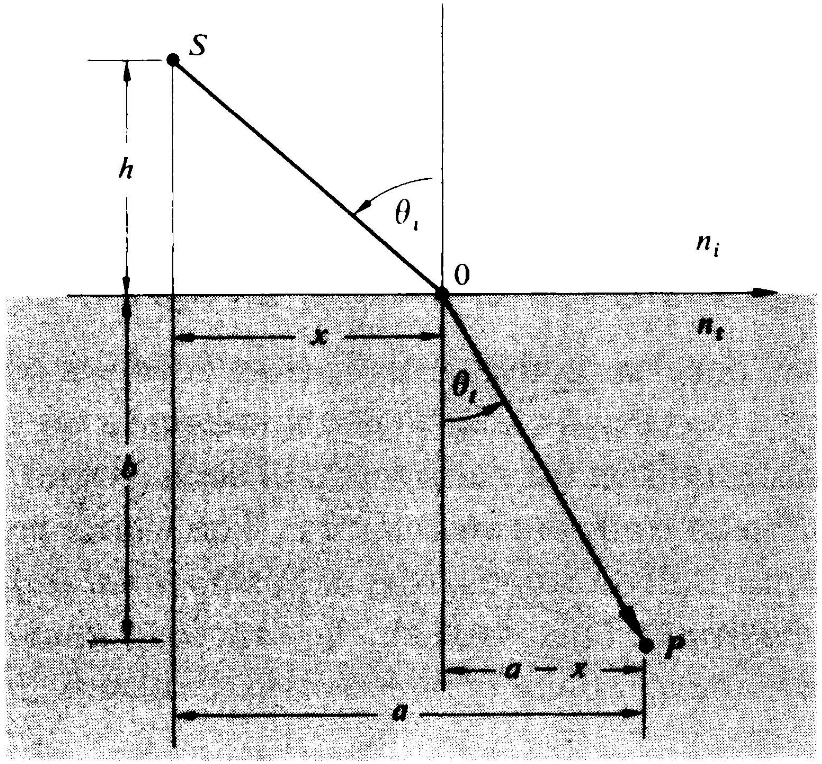
According to Fermat's principle:

$$t = \frac{\overline{SO}}{v_i} + \frac{\overline{OP}}{v_t}$$

The solution:

$$\frac{\sin \theta_i}{v_i} = \frac{\sin \theta_t}{v_t}$$

The law of refraction



Optical path length

The transit time from S to P:

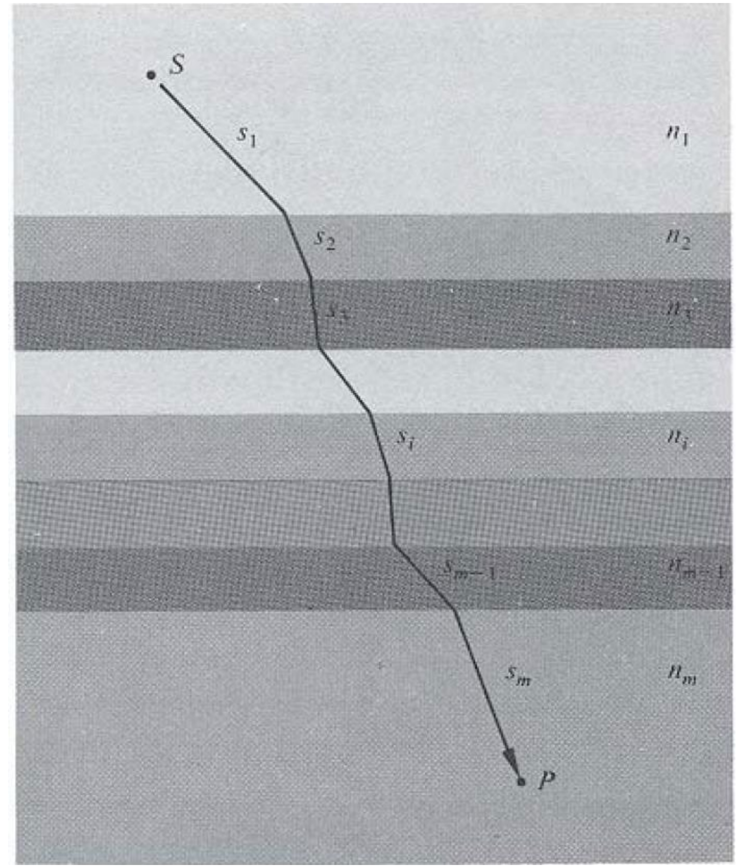
$$t = \frac{s_1}{v_1} + \frac{s_2}{v_2} + \cdots + \frac{s_m}{v_m}$$

or

$$t = \sum_{i=1}^{m} \frac{s_i}{v_i} = \frac{1}{c} \sum_{i=1}^{m} n_i s_i$$

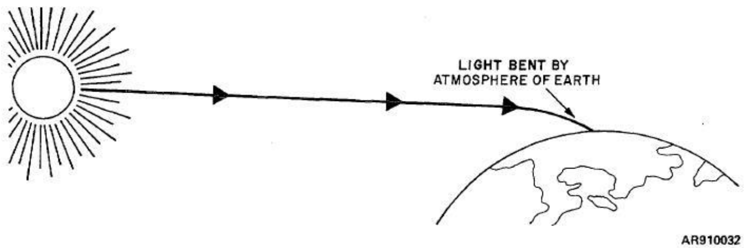
For an inhomogeneous medium, **the optical path length**:

$$OPL = \int_S^P n(s)\, ds$$

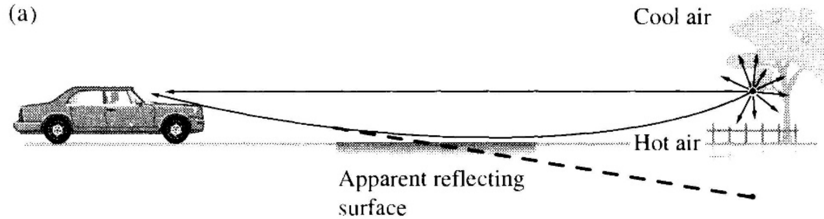


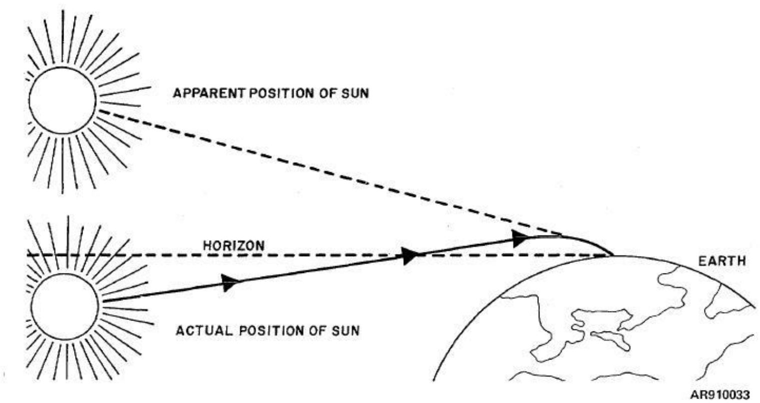
- It corresponds to the distance in vacuum equivalent to the distance traversed (s) in the medium of index n.
- Fermat’s Principle: light, in going from point S to P, traverses the route having the smallest optical length.

Fermat and mirages

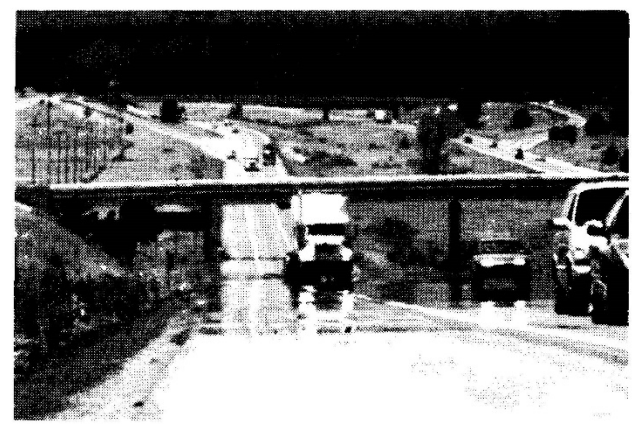


(a)





(b)



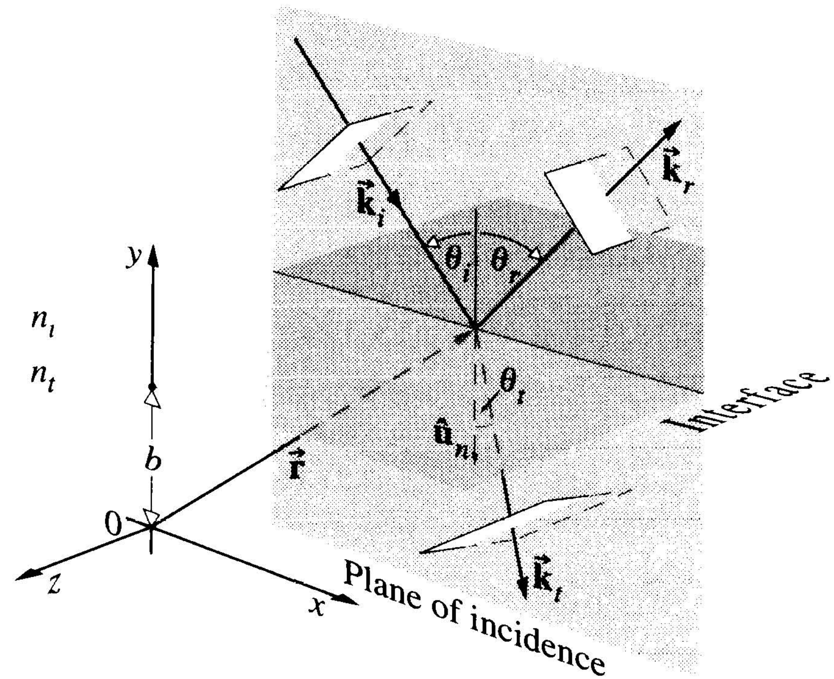
3.6 The electromagnetic approach

- ❑ As to reflection and refraction phenomena:
- ✓ Scattering theory
- ✓ Fermat's principle
- ❑ Another powerful approach:
- ✓ **Electromagnetic theory**

3.6.1 Waves at an interface

Suppose that the incident monochromatic lightwave is planar:

$$\vec{E}_i = \vec{E}_{0i} \cos\left(\vec{k}_i \cdot \vec{r} - \omega_i t\right)$$



Reflected lightwave:

$$\vec{E}_r = \vec{E}_{0r} \cos\left(\vec{k}_r \cdot \vec{r} - \omega_r t + \epsilon_r\right)$$

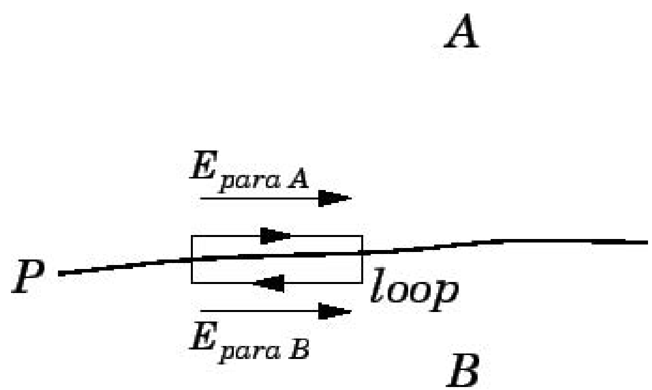
Refracted lightwave:

$$\vec{E}_t = \vec{E}_{0t} \cos\left(\vec{k}_t \cdot \vec{r} - \omega_t t + \epsilon_t\right)$$

ϵ_r and ϵ_t: phase constants relative to E_i

Start from Faraday's induction law:

$$\oint_C \vec{E} \cdot d\ell = -\iint_A \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

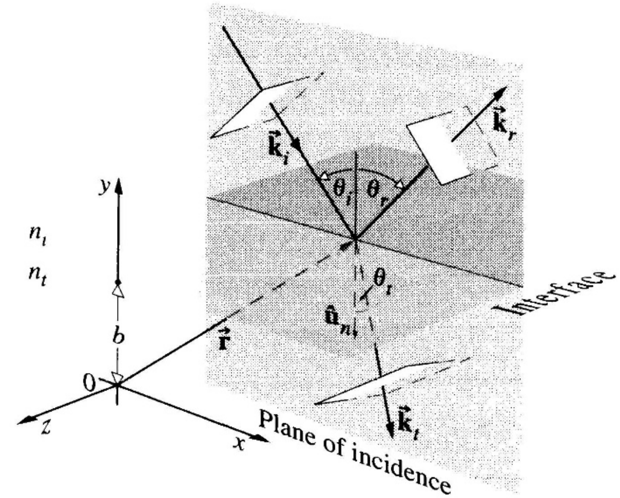


$$E_{\parallel A} = E_{\parallel B}$$

- Conclusion (Boundary conditions): continuity of tangential E

Boundary condition: continuity of tangential E

$\hat{u}_n$: the unit vector normal to the interface



$$\hat{u}_n \times \vec{E}_i + \hat{u}_n \times \vec{E}_r = \hat{u}_n \times \vec{E}_t$$

$$\hat{u}_n \times \vec{E}_{0i} \cos\left(\vec{k}_i \cdot \vec{r} - \omega_i t\right)$$
$$+\hat{u}_n \times \vec{E}_{0r} \cos\left(\vec{k}_r \cdot \vec{r} - \omega_r t + \epsilon_r\right)$$
$$= \hat{u}_n \times \vec{E}_{0t} \cos\left(\vec{k}_t \cdot \vec{r} - \omega_t t + \epsilon_t\right)$$

This relationship must obtain at any instant in time and at any point on the interface. Three electric fields have the same functional dependence on the t and r.

$$\left(\vec{k}_i \cdot \vec{r} - \omega_i t\right)\Big|_{y=b} = \left(\vec{k}_r \cdot \vec{r} - \omega_r t + \epsilon_r\right)\Big|_{y=b} = \left(\vec{k}_t \cdot \vec{r} - \omega_t t + \epsilon_t\right)\Big|_{y=b}$$

The frequency is unchanged under linear region

$$\left(\vec{k}_i \cdot \vec{r}\right)\Big|_{y=b} = \left(\vec{k}_r \cdot \vec{r} + \epsilon_r\right)\Big|_{y=b} = \left(\vec{k}_t \cdot \vec{r} + \epsilon_t\right)\Big|_{y=b}$$

$$\left(\vec{k}_i \cdot \vec{r}\right)\Big|_{\,y=b} = \left(\vec{k}_r \cdot \vec{r} + \epsilon_r\right)\Big|_{\,y=b}$$

$$\left[\left(\vec{k}_i - \vec{k}_r\right) \cdot \vec{r}\right]\Big|_{\,y=b} = \epsilon_r$$

The endpoint of *r* sweeps out a plane perpendicular to the vector $\vec{k}_i - \vec{k}_r$

And the vector $\vec{k}_i - \vec{k}_r$ is parallel to $\hat{u}_n$

$$\hat{u}_n \times \left(\vec{k}_i - \vec{k}_r\right) = 0$$

The law of reflection

$$k_i \sin\theta_i = k_i \sin\theta_r$$

$$\theta_i = \theta_r$$

$$\left(\vec{k}_i \cdot \vec{r}\right)\Big|_{\,y=b} = \left(\vec{k}_t \cdot \vec{r} + \epsilon_t\right)\Big|_{\,y=b}$$

Snell's law

similarly $k_i \sin\theta_i = k_t \sin\theta_t$

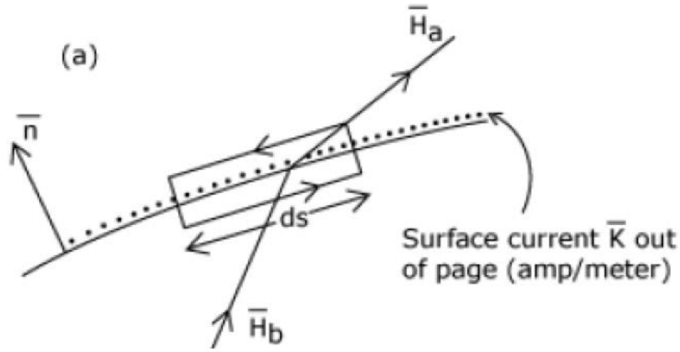
$$n_i \sin\theta_i = n_t \sin\theta_t$$

Boundary condition: continuity of tangential H

Ampere-Maxwell's Law

$$\oint_C \vec{B} \cdot d\ell = \mu_0 \iint_A \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{S}$$

Ampere-Maxwell's Law in matter



$$\oint_C \vec{H} \cdot d\ell = \iint_A \left(\vec{J}_f + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{S}$$

$$\vec{H} = \vec{B}/\mu$$

$$\vec{D} = \epsilon \vec{E}$$

Dielectric-free space boundary condition

$$H_{\parallel A} = H_{\parallel B}$$

3.6.2 The Fresnel Equations

Case 1: $\vec{E}$ perpendicular to the plane-of-incidence

$$\vec{k} \times \vec{E} = v\vec{B}$$

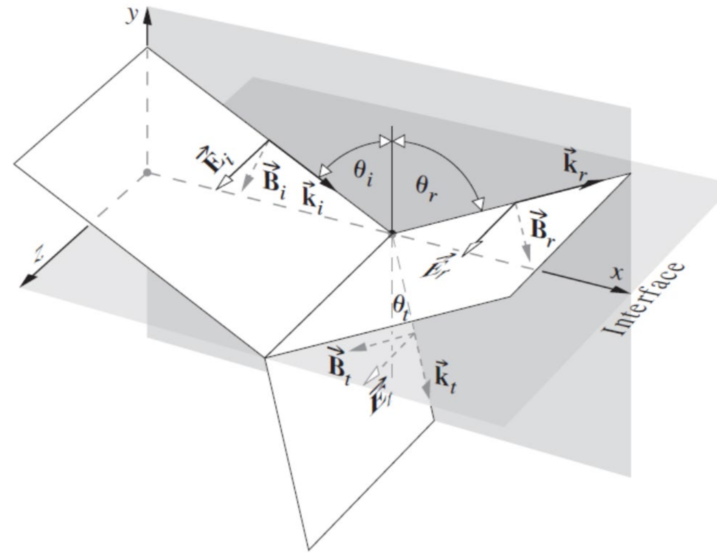
$$\vec{k} \cdot \vec{E} = 0$$

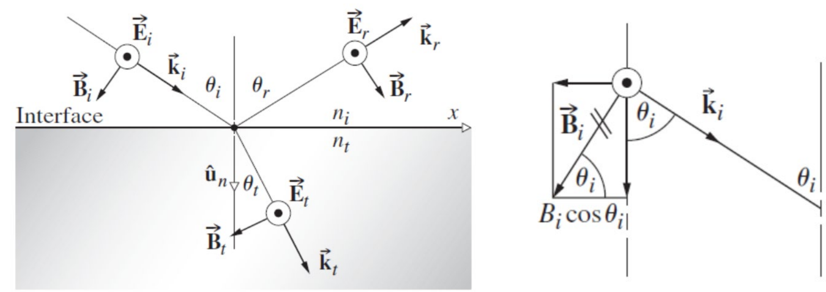
Continuity of the tangential components of the $\vec{E}$-field at the boundary

$$E_{0i} + E_{0r} = E_{0t}$$

Continuity of the tangential components of the $\vec{B}/\mu$ at the boundary

$$-\frac{B_i}{\mu_i}\cos\theta_i + \frac{B_r}{\mu_i}\cos\theta_r = -\frac{B_t}{\mu_t}\cos\theta_t$$





Reflection and refraction of an incoming wave whose $\vec{E}$-field is normal to the plane of incidence.

Fresnel equation (for any linear, isotropic, homogeneous dielectrics)

Amplitude reflection coefficient:

$$r_\perp \equiv \left(\frac{E_{0r}}{E_{0i}}\right)_\perp = \frac{n_i \cos\theta_i - n_t \cos\theta_t}{n_i \cos\theta_i + n_t \cos\theta_t}$$

Amplitude transmission coefficient:

$$t_\perp \equiv \left(\frac{E_{0t}}{E_{0i}}\right)_\perp = \frac{2n_i \cos\theta_i}{n_i \cos\theta_i + n_t \cos\theta_t}$$

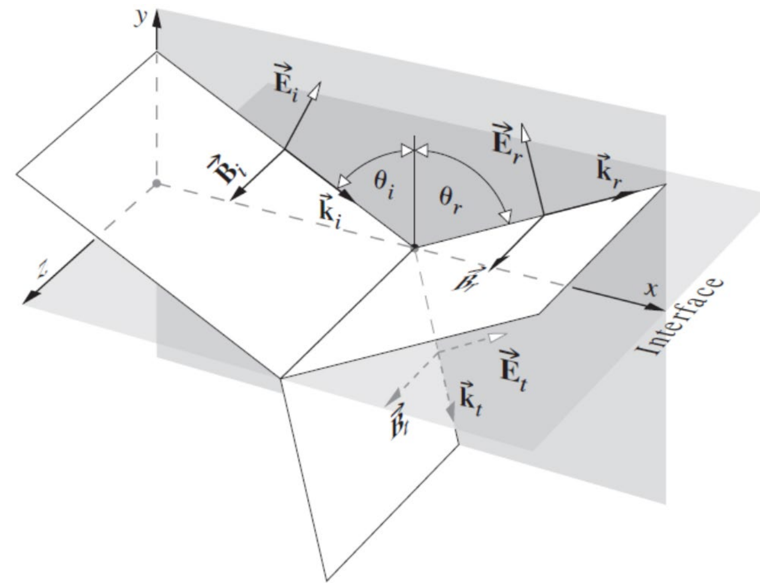
Case 2: $\vec{E}$ parallel to the plane-of-incidence

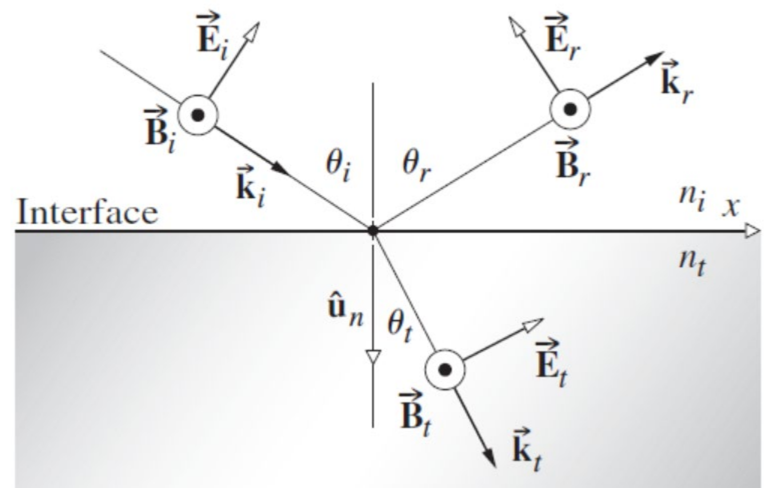
Continuity of the tangential components of the $\vec{E}$-field at the boundary

$$E_{0i}\cos\theta_i - E_{0r}\cos\theta_r = E_{0t}\cos\theta_t$$

Continuity of the tangential components of the $\vec{B}/\mu$ at the boundary

$$\frac{B_i}{\mu_i} + \frac{B_r}{\mu_i} = \frac{B_t}{\mu_t}$$





Fresnel equation (for any linear, isotropic, homogeneous dielectrics)

Amplitude reflection coefficient:

$$r_{\parallel} \equiv \left(\frac{E_{0r}}{E_{0i}}\right)_{\parallel} = \frac{n_t \cos\theta_i - n_i \cos\theta_t}{n_i \cos\theta_t + n_t \cos\theta_i}$$

$$r_{\perp} \equiv \left(\frac{E_{0r}}{E_{0i}}\right)_{\perp} = \frac{n_i \cos\theta_i - n_t \cos\theta_t}{n_i \cos\theta_i + n_t \cos\theta_t}$$

Amplitude transmission coefficient:

$$t_{\parallel} \equiv \left(\frac{E_{0t}}{E_{0i}}\right)_{\parallel} = \frac{2n_i \cos\theta_i}{n_i \cos\theta_t + n_t \cos\theta_i}$$

$$t_{\perp} \equiv \left(\frac{E_{0t}}{E_{0i}}\right)_{\perp} = \frac{2n_i \cos\theta_i}{n_i \cos\theta_i + n_t \cos\theta_t}$$

Further simplification using Snell's law

$$r_{\parallel} = +\frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)}$$

$$r_{\perp} = -\frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)}$$

$$t_{\parallel} = +\frac{2\sin\theta_t \cos\theta_i}{\sin(\theta_i + \theta_t)\cos(\theta_i - \theta_t)}$$

$$t_{\perp} = +\frac{2\sin\theta_t \cos\theta_i}{\sin(\theta_i + \theta_t)}$$

Homework

Derive **Fresnel equations**

3.6.3 Interpretation of the Fresnel equations

Amplitude coefficients under nearly normal incidence

$$\left[r_{\parallel}\right]_{\theta_i=0} = -[r_{\perp}]_{\theta_i=0} = \left[\frac{n_t - n_i}{n_i + n_t}\right]_{\theta_i=0}$$

$$\left[t_{\parallel}\right]_{\theta_i=0} = [t_{\perp}]_{\theta_i=0} = \left[\frac{2n_i}{n_i + n_t}\right]_{\theta_i=0}$$

Example: at an air ($n_i = 1$)–glass ($n_t = 1.5$) interface

$$\left[r_{\parallel}\right]_{\theta_i=0} = -[r_{\perp}]_{\theta_i=0} = \left[\frac{n_t - n_i}{n_i + n_t}\right]_{\theta_i=0} = 0.2$$

External Reflection ($n_t > n_i$)

$$r_\parallel = +\frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)}$$

When $\theta_i + \theta_t = 90°$, $r_\parallel = 0$

This particular value of the incident angle is referred to as **Polarization angle** (θ_p). The phase shift 180° happens at θ_p.

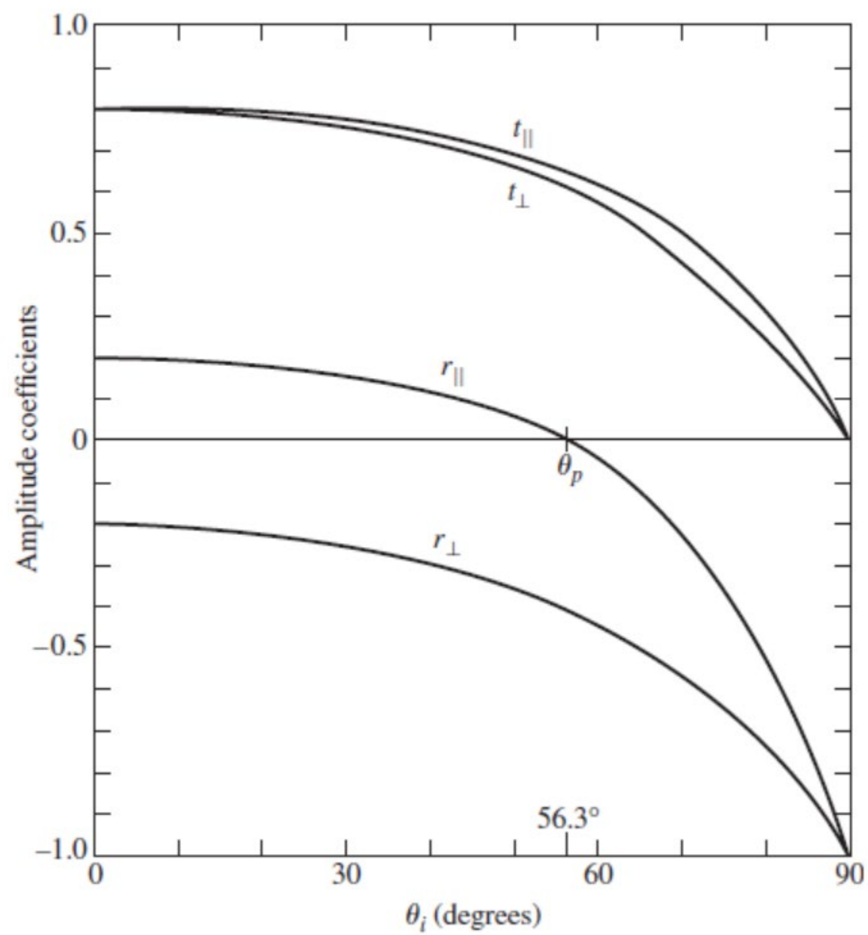
$$r_\perp = -\frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)} < 0$$

for all incident angle

When $\theta_i = 90°$ (**glancing incidence**),

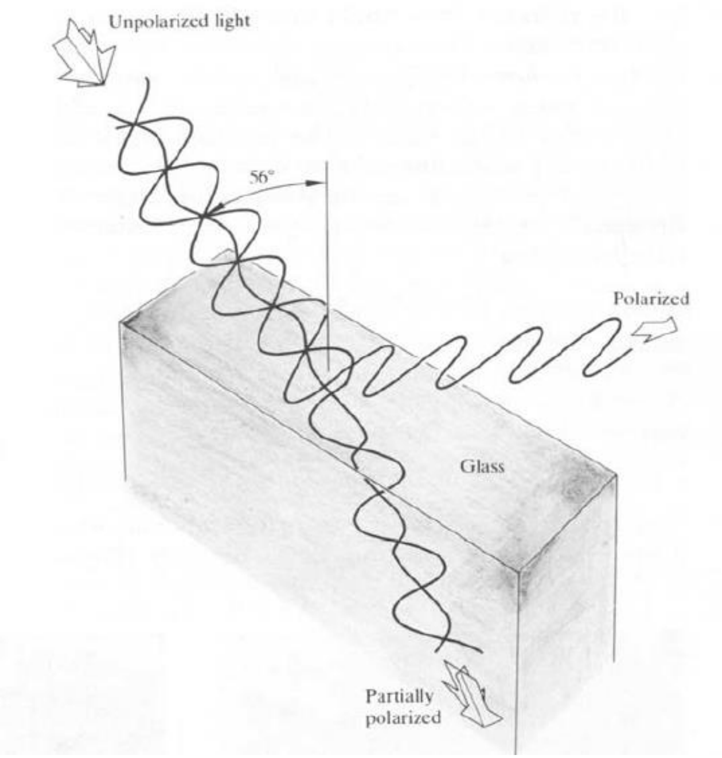
$$r_\parallel = \frac{n_t \cos\theta_i - n_i \cos\theta_t}{n_i \cos\theta_t + n_t \cos\theta_i} = -1$$

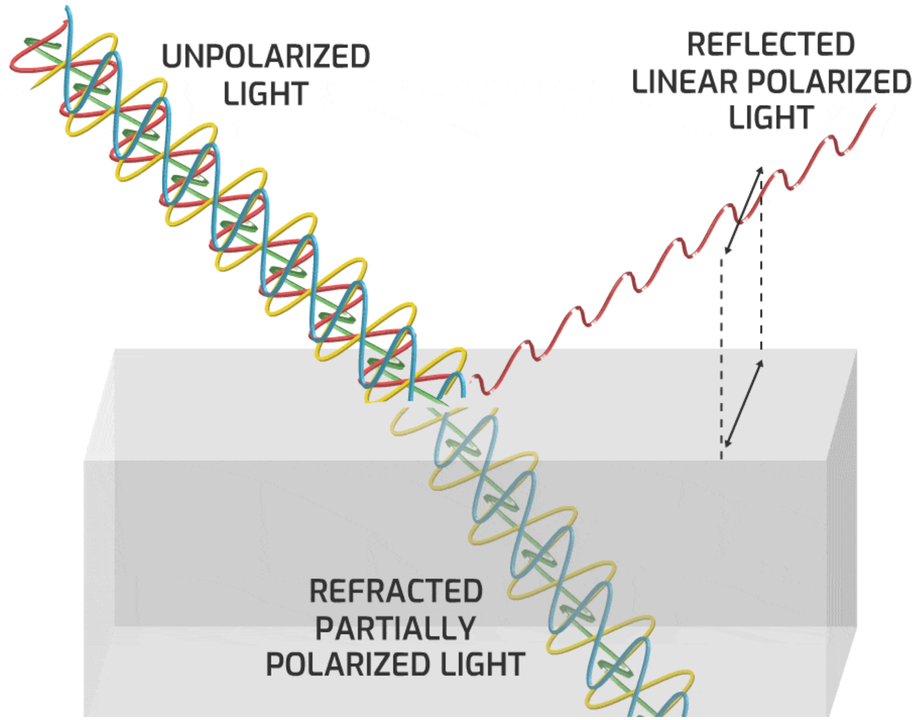
$$r_\perp = \frac{n_i \cos\theta_i - n_t \cos\theta_t}{n_i \cos\theta_i + n_t \cos\theta_t} = -1$$



The amplitude coefficients of the reflection and transmission as a function of incident angle. These correspond to external reflection n_t>n_i at an air-glass interface.

Polarization angle





Internal Reflection ($n_i > n_t$)

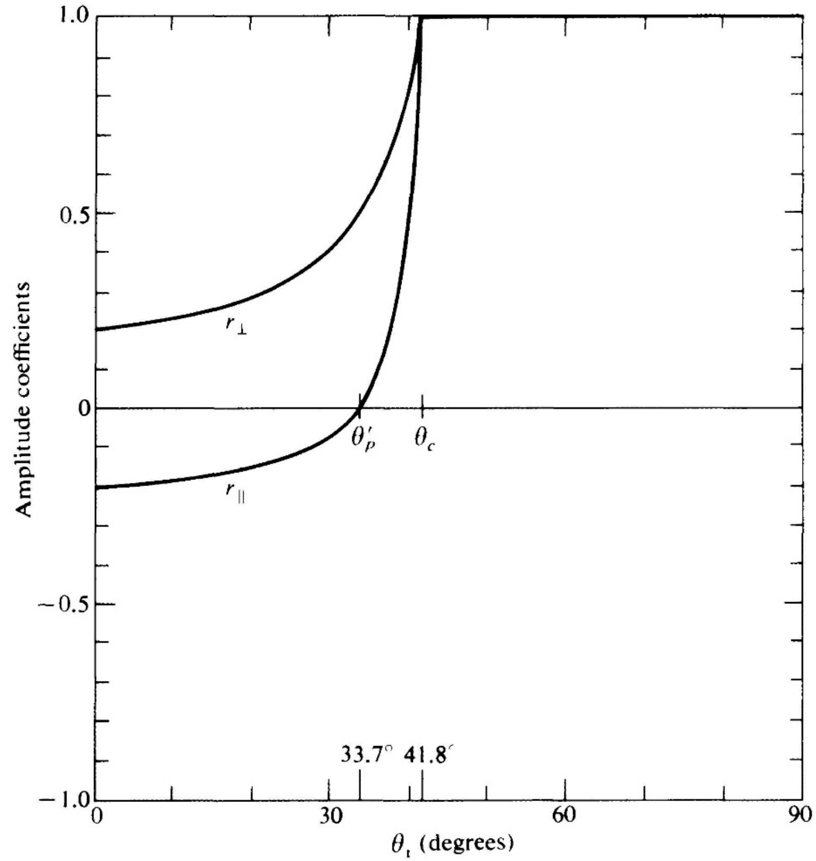
Nearly normal incidence

$$[r_\parallel]_{\theta_i=0} = -[r_\perp]_{\theta_i=0} = \left[\frac{n_t - n_i}{n_i + n_t}\right]_{\theta_i=0}$$

With increasing θ_i

$$r_\perp = -\frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)}$$ Positive

$$r_\parallel = +\frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)}$$ From negative to positive at $\theta_i + \theta_t = 90°$



The amplitude coefficients of reflection as a function of incident angle. These correspond to internal reflection at an air-glass interface.

Passing through zero at the polarization angle θ_p'

When $\theta_t = 90°$ $$r_\parallel = \frac{n_t \cos\theta_i - n_i \cos\theta_t}{n_i \cos\theta_t + n_t \cos\theta_i} = 1$$

$$r_\perp = \frac{n_i \cos\theta_i - n_t \cos\theta_t}{n_i \cos\theta_i + n_t \cos\theta_t} = 1$$

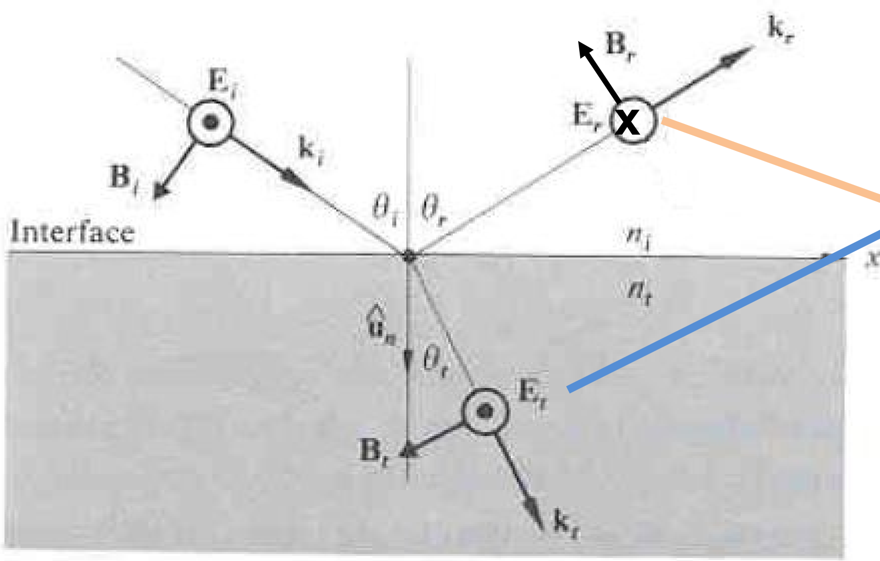
Phase shift for external reflection

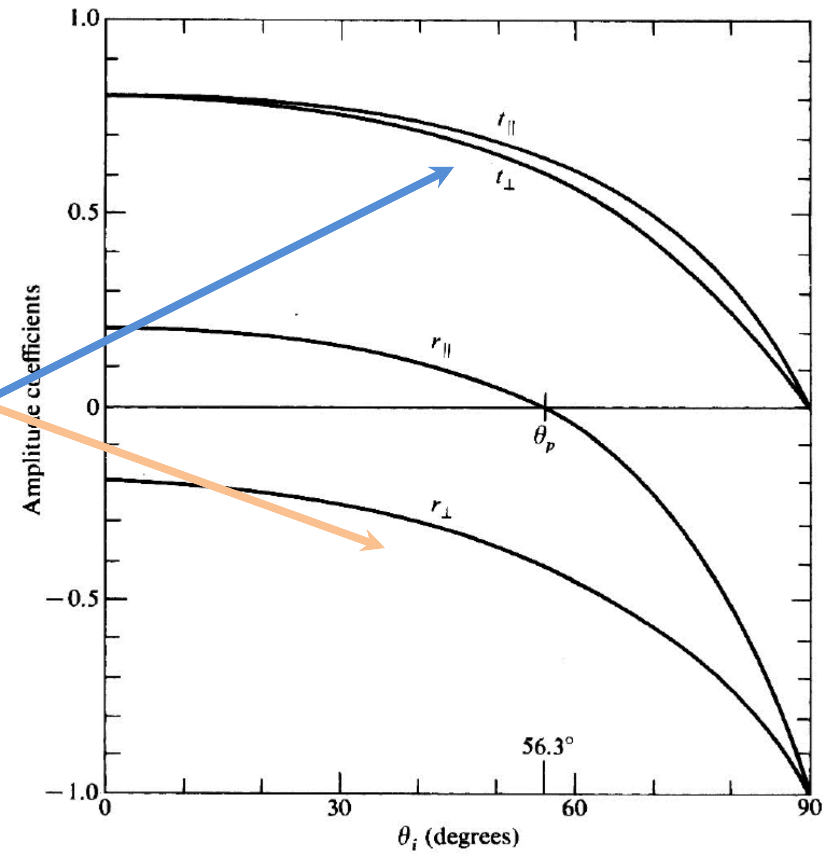
$\vec{E}$ perpendicular to the plane-of-incidence

$$r_{\parallel} = +\frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)} \qquad r_{\perp} = -\frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)}$$

$$t_{\parallel} = +\frac{2\sin\theta_t\cos\theta_i}{\sin(\theta_i + \theta_t)\cos(\theta_i - \theta_t)}$$

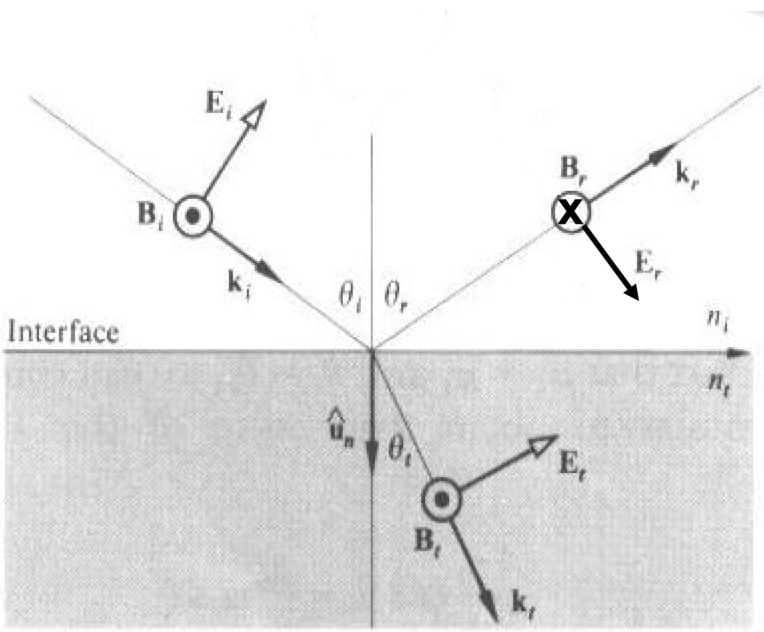
$$t_{\perp} = +\frac{2\sin\theta_t\cos\theta_i}{\sin(\theta_i + \theta_t)}$$

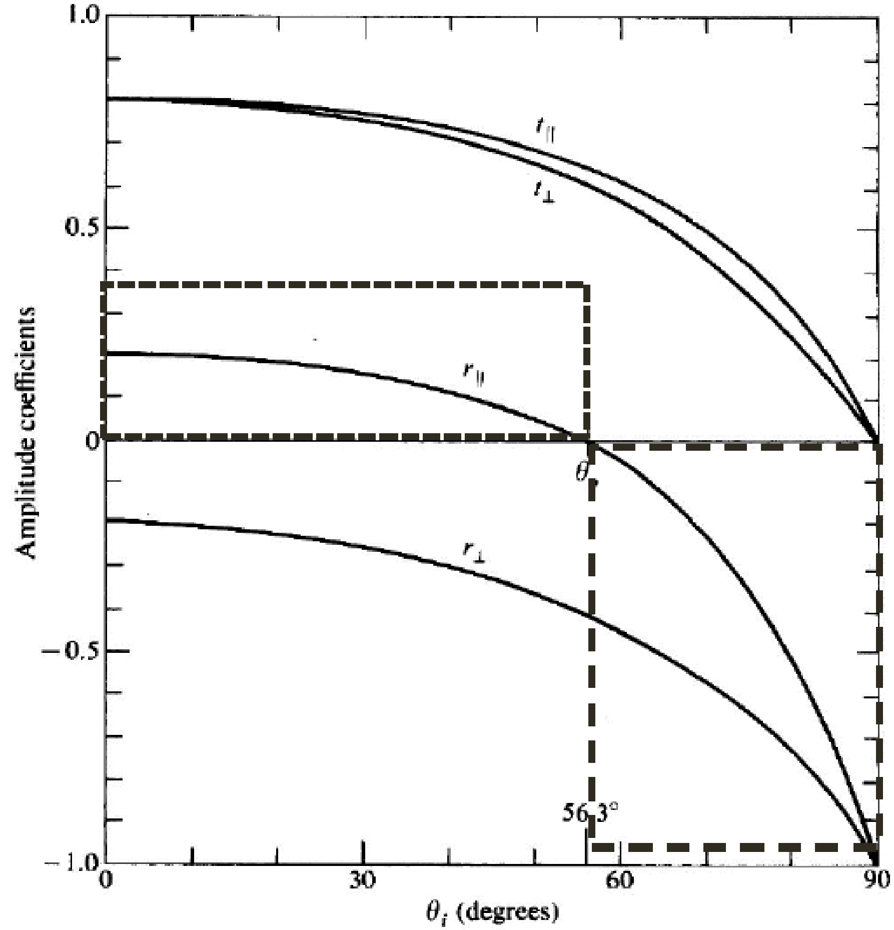




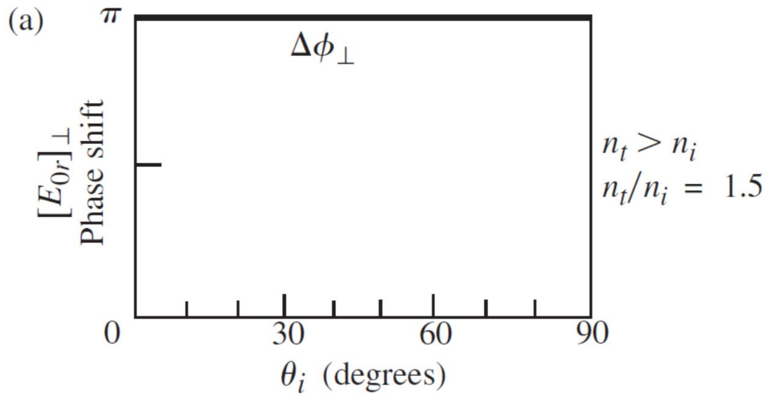
Phase shift for external reflection

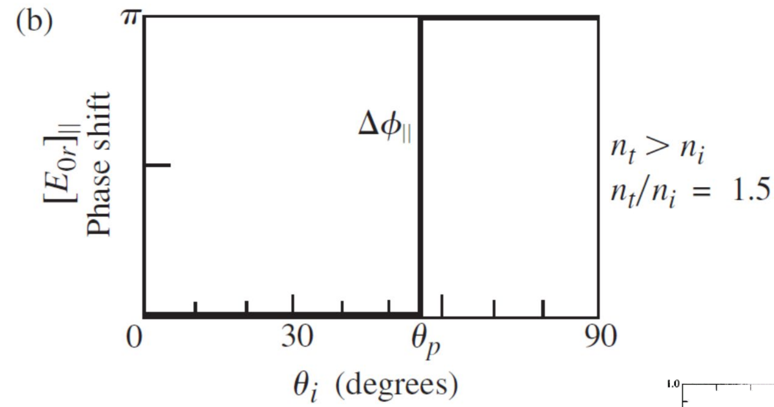
$\vec{E}$ **parallel to the plane-of-incidence**

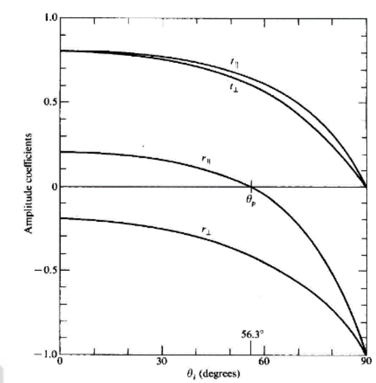


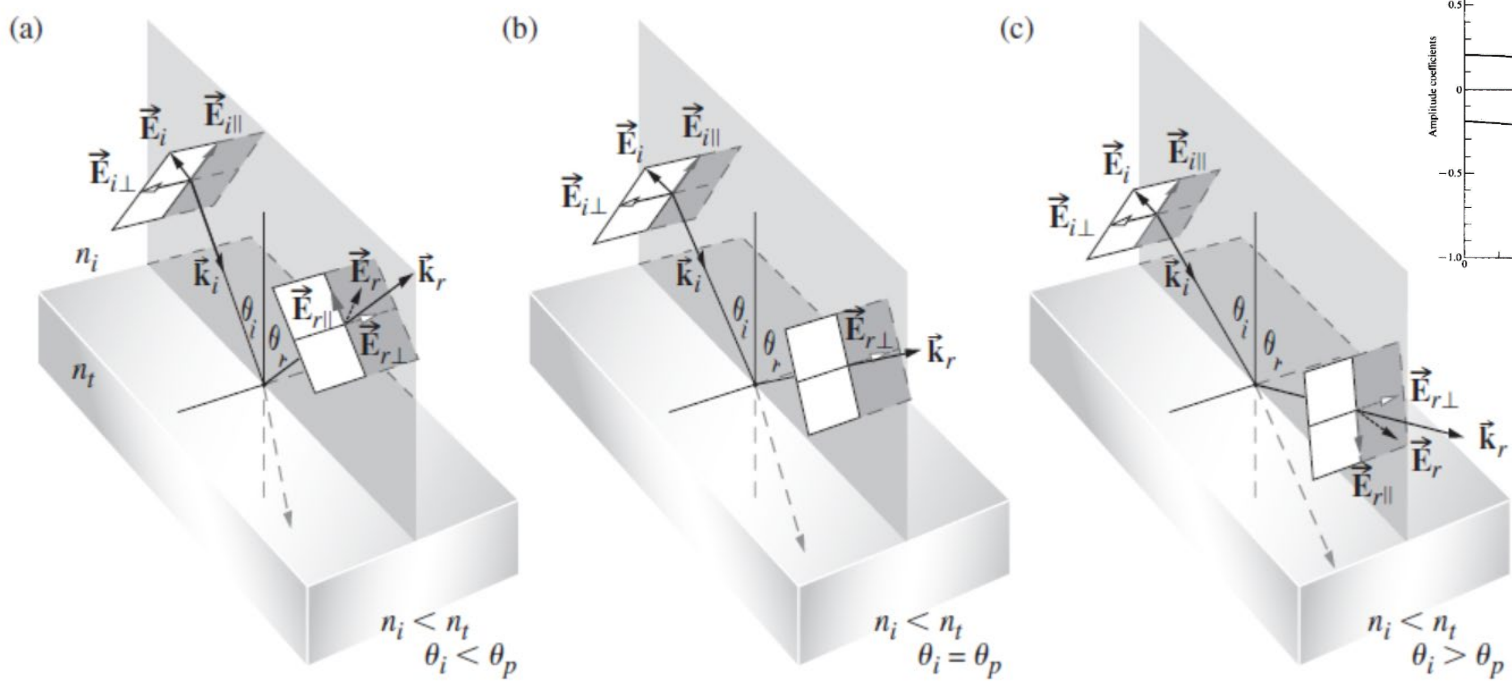


Phase shift for external reflection

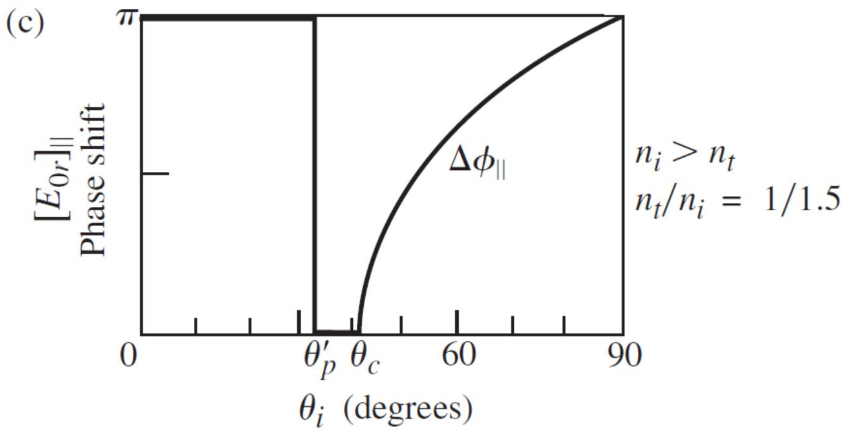


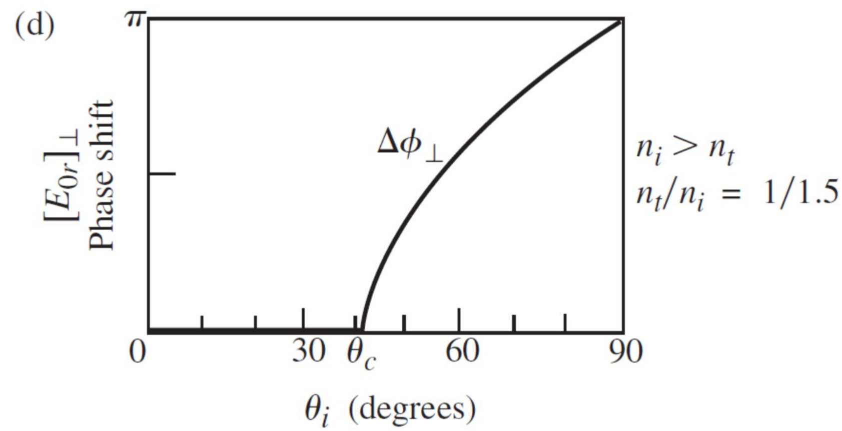


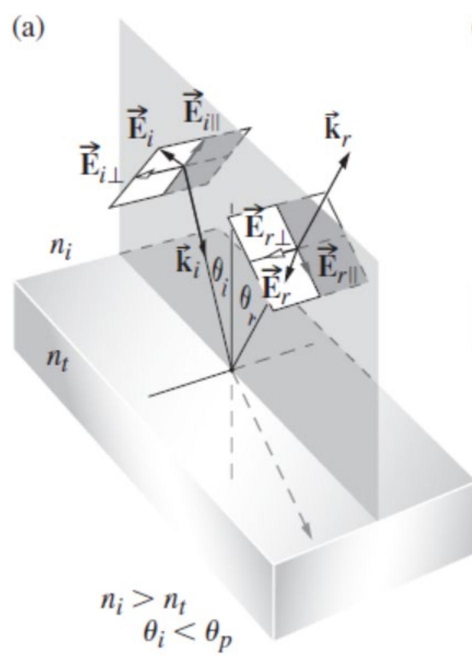


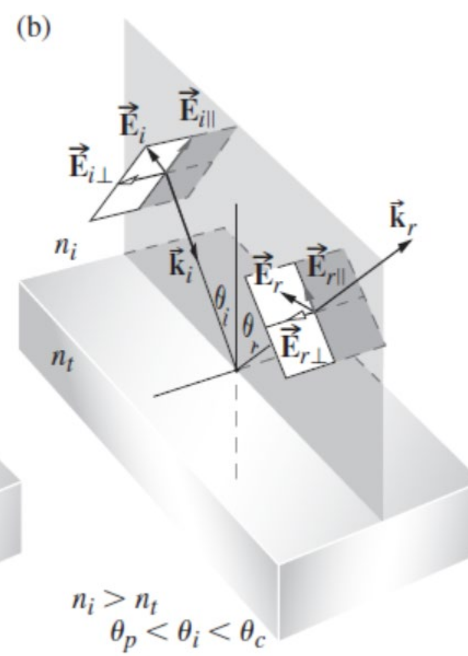


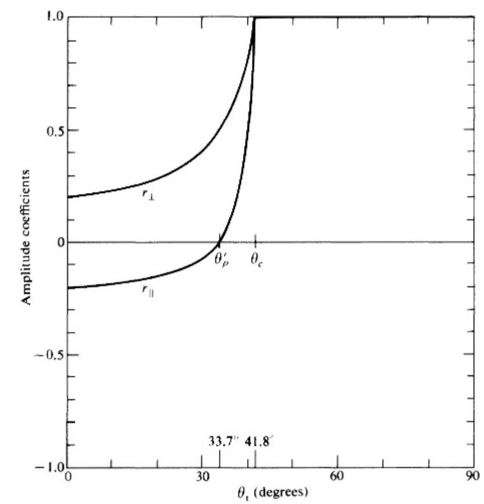
Phase shift for internal reflection











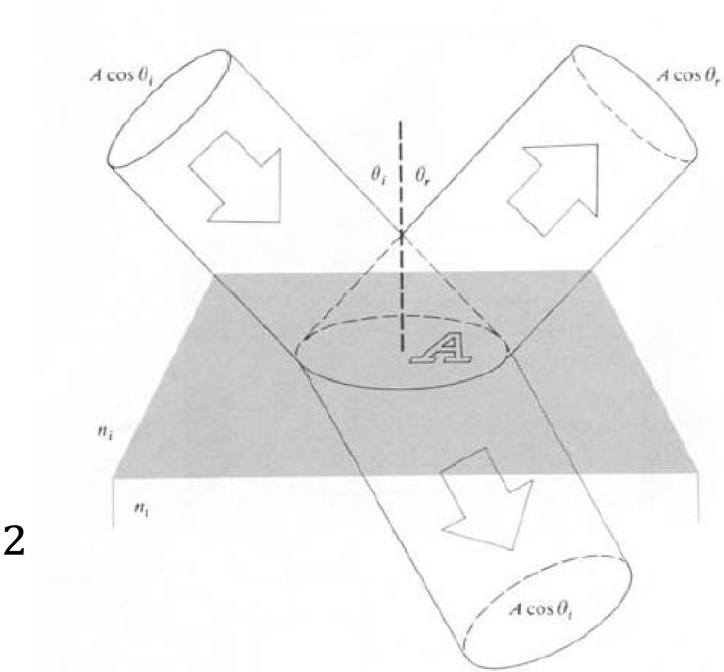
Reflectance and transmittance

Poynting vector: the power per unit area crossing a surface in vacuum

$$\vec{S} = c^2 \epsilon_0 \vec{E} \times \vec{B}$$

Radiant flux density (W/m^2): average energy per unit time crossing a unit area normal to $\vec{S}$.

$$I = \langle S \rangle_T = \frac{\boxed{c\epsilon_0} E_0{}^2}{2}$$



Reflectance (R):

$$R \equiv \frac{I_r A \cos\theta_r}{I_i A \cos\theta_i} = \frac{I_r}{I_i} = \left(\frac{E_{0r}}{E_{0i}}\right)^2 = r^2$$

Transmittance (T):

$$T \equiv \frac{I_t A \cos\theta_t}{I_i A \cos\theta_i} = \frac{n_t \cos\theta_t}{n_i \cos\theta_i}\left(\frac{E_{0t}}{E_{0i}}\right)^2 = \left(\frac{n_t \cos\theta_t}{n_i \cos\theta_i}\right) t^2$$

The reason that T is not simply equal to t^2: 1) $I \propto v$; 2) the cross-sectional areas of the incident and refracted beams are different

Conservation of energy: the total energy flowing into area A per unit time must equal the energy flowing outward from it per unit time

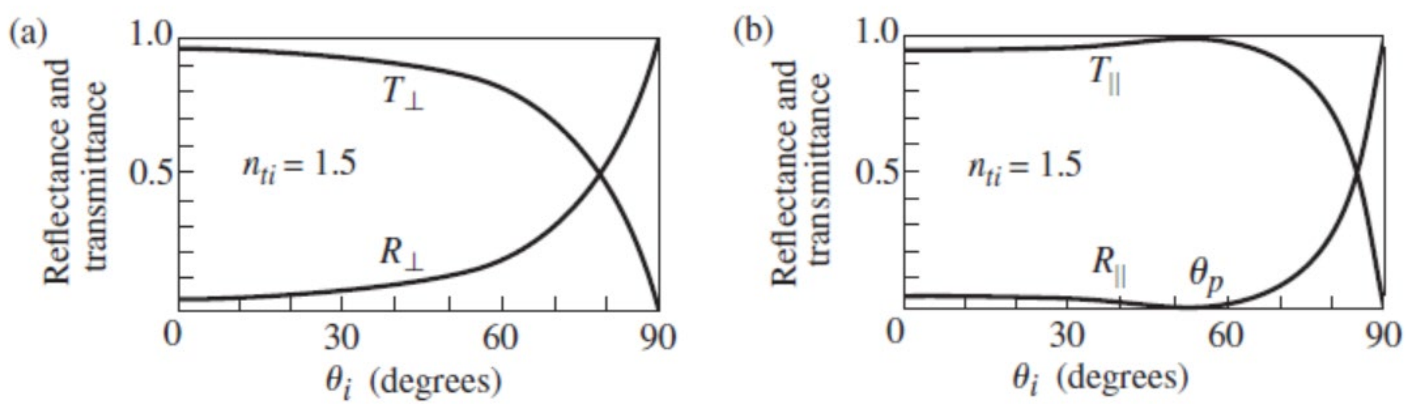
$$I_i A \cos\theta_i = I_r A \cos\theta_r + I_t A \cos\theta_t$$

$$R + T = 1$$

The component forms:

$$R_\perp = {r_\perp}^2 \qquad T_\perp = \left(\frac{n_t \cos\theta_t}{n_i \cos\theta_i}\right){t_\perp}^2$$

$$R_\parallel = {r_\parallel}^2 \qquad T_\parallel = \left(\frac{n_t \cos\theta_t}{n_i \cos\theta_i}\right){t_\parallel}^2$$



Normal incidence

$$R = R_{\parallel} = R_{\perp} = \left(\frac{n_t - n_i}{n_t + n_i}\right)^2$$

$$T = T_{\parallel} = T_{\perp} = \frac{4n_t n_i}{(n_t + n_i)^2}$$

4% of the light incident normally on an air-glass interface will be reflected back.

Because it's a lot brighter outside than inside the building,
you have no trouble seeing outside from inside